WIKIPEDIA 25
25 years of the free encyclopedia

Oryzomys albiventer

Oryzomys albiventer, commonly known as the **White-bellied Rice Rat**, is a rodent in the genus *Oryzomys* of family Cricetidae from interior western Mexico, in the states of Jalisco, Guanajuato, and Michoacán. First described in 1901 as a separate species, it was later lumped under *O. couesi* and the marsh rice rat (*O. palustris*) until it was reinstated as a species in 2009. It differs from neighboring *Oryzomys* populations in size and measurements and is a large, brightly colored species with a long tail and robust skull and molars. Its range has been much impacted by agricultural development, but isolated populations are thought to persist.

Oryzomys albiventer



Drawing of *Oryzomys molestus*,[1] a synonym of *Oryzomys albiventer*.[2]

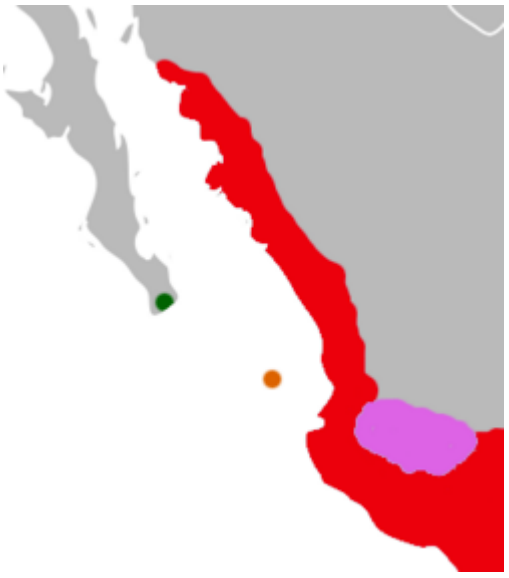
Scientific classification

Kingdom:	Animalia
Phylum:	Chordata
Class:	Mammalia
Order:	Rodentia
Family:	Cricetidae
Subfamily:	Sigmodontinae
Genus:	*Oryzomys*
Species:	***O. albiventer***

Binomial name

Oryzomys albiventer

Merriam, 1901



Taxonomy

Oryzomys albiventer was first described by C.H. Merriam in 1901 on the basis of ten specimens from Ameca, Jalisco.[8] He named the animal *albiventer* after the white color of its underparts[9] and considered it most closely related to *Oryzomys aquaticus*[4] (currently included in *Oryzomys couesi*).[10] Two years later, D.G. Eliot described *Oryzomys molestus* on the basis of a single individual from Ocotlán, Jalisco; the name *molestus* means "troublesome, irksome". Eliot considered *Oryzomys fulgens*,[5] another current synonym of *O. couesi*,[10] as the closest relative of his new species.[5] In his 1918 review of North American *Oryzomys*, E.A. Goldman assessed the holotype of *O. molestus*, an old male, as merely a large example of *albiventer*, and reduced *albiventer* to one of many subspecies of *O. couesi*.[6] He considered it closely related to three other highland Mexican forms.[11] In 1960, E.R. Hall argued that *O. couesi* was the same species as the marsh rice rat (*O. palustris*) of the United States, and listed *albiventer* as a subspecies of the latter.[7] Later, *O. couesi* was again accepted as separate from the marsh rice rat, but *O. albiventer* was still classified under *O. couesi*.[12]

In a 2009 revision of the *Oryzomys* of western Mexico, M.D. Carleton and J. Arroyo-Cabrales noted substantial differences in coloration and measurements between highland populations (*albiventer*) and lowland populations

(*mexicanus*) in Jalisco.[13] On the basis of these differences, they recognized *O. albiventer* as a species separate from the lowland populations, which they classified under *O. couesi*.[14] They confirmed that Eliot's *molestus* was based on a large example of *O. abiventer*,[9] but left the status of the three forms Goldman had associated with *albiventer*—*crinitus*, *aztecus*, and *regillus*—open, noting that there was no convincing evidence that these represented the same species as *O. albiventer*.[15] The identity and exact provenance of *fulgens* (supposed to be from the Valley of Mexico), and consequently its relationship to *O. albiventer*, remain unknown.[16]

Distribution of *Oryzomys albiventer* (in pink) and other western Mexican *Oryzomys*.

Synonyms[3]

- *Oryzomys albiventer* Merriam, 1901[4]
- *Oryzomys molestus* Eliot, 1903[5]
- *Oryzomys couesi albiventer*: Goldman, 1918[6]
- *Oryzomys palustris albiventer*: Hall, 1960[7]

O. albiventer is part of the genus *Oryzomys*, which currently includes about eight species distributed from the eastern United States (*O. palustris*) into northwestern South America (*O. gorgasi*).[17] *O. albiventer* is further part of the *O. couesi* section, which is centered on the widespread Central American *O. couesi* and also includes various other species with more limited and peripheral distributions.[2] Many aspects of the systematics of the *O. couesi* section remain unclear and it is likely that the current classification underestimates the true diversity of the group.[18] *Oryzomys* previously included many other species, which were progressively removed in various studies culminating in a contribution by Marcelo Weksler and coworkers in 2006 that removed more than forty species from the genus.[19] All are classified in the tribe Oryzomyini ("rice rats"), a diverse assemblage of American rodents of over a hundred species,[20] and on higher taxonomic levels in the subfamily Sigmodontinae of family Cricetidae, along with hundreds of other species of mainly small rodents.[21]

In 1904, Eliot used the common name "white-bellied rice rat" for *O. albiventer*[22] and "Ocotlan rice rat" for *O. molestus*.[1] In 1918, Goldman also used "white-bellied rice rat" for *O. albiventer*.[6]

Description

Oryzomys albiventer is a large and long-tailed *Oryzomys*.[9] The upperparts are brightly ochraceous, becoming grayer toward the front.[4] The hairs on the underparts are pale gray near the bases and white in the outer half, so that the underparts appear pale grayish according to Carleton and Arroyo-Cabrales (not white as claimed by Merriman).[9] The tail is dark above and light below.[4] The skull and molars are relatively robust.[9] *O. albiventer* has broad zygomatic arches (cheekbones), long incisive foramina (perforations of the palate between the incisors and the molars), and long nasal bones that extend behind the premaxillary bones.[4] Compared to its lowland relative *O. couesi mexicanus*, *O. albiventer* is larger and more brightly colored and has larger molars but narrower incisive foramina.[23]

In twelve specimens, total length is 245 to 314 mm (9.6 to 12.4 in), averaging 285.4 mm (11.24 in); head and body length is 116 to 142 mm (4.6 to 5.6 in), averaging 130.0 mm (5.12 in); tail length is 129 to 173 mm (5.1 to 6.8 in), averaging 155.4 mm (6.12 in); hindfoot length is 33 to 40 mm (1.3 to 1.6 in), averaging 36.1 mm (1.42 in); and skull length (occipitonasal length) is 30.0 to 34.5 mm (1.18 to 1.36 in), averaging 32.9 mm (1.30 in).[24]

Distribution and conservation

O. albiventer occurs at about 1,200 to 1,800 m (3,900 to 5,900 ft) elevation in northern Michoacán, southern Guanajuato, and central and eastern Jalisco,[9] mostly in the area around Lake Chapala. Its range has seen massive agricultural development and although populations may survive, the current distribution of the species is certainly highly fragmented. More survey work is needed to assess the distribution and status of *O. albiventer*.[16]

References

1. Eliot, 1904, p. 240
2. Carleton and Arroyo-Cabrales, 2009, p. 117
3. Carleton and Arroyo-Cabrales, 2009, pp. 117–118
4. Merriam, 1901, p. 279
5. Eliot, 1903, p. 145
6. Goldman, 1918, p. 38
7. Hall, 1960, p. 173
8. Merriam, 1901, pp. 279–280
9. Carleton and Arroyo-Cabrales, 2009, p. 118
10. Carleton and Arroyo-Cabrales, 2006, p. 116
11. Goldman, 1918, p. 37
12. Carleton and Arroyo-Cabrales, 2009, p. 95
13. Carleton and Arroyo-Cabrales, 2009, p. 99
14. Carleton and Arroyo-Cabrales, 2009, pp. 104, 106
15. Carleton and Arroyo-Cabrales, 2009, p. 113
16. Carleton and Arroyo-Cabrales, 2009, p. 115
17. Carleton and Arroyo-Cabrales, 2009, p. 106
18. Carleton and Arroyo-Cabrales, 2009, p. 107
19. Weksler et al., 2006, table 1
20. Weksler, 2006, p. 3
21. Musser and Carleton, 2005
22. Eliot, 1904, p. 236
23. Carleton and Arroyo-Cabrales, 2009, pp. 101–102
24. Carleton and Arroyo-Cabrales, 2009, table 2

Literature cited

- Carleton, M.D. and Arroyo-Cabrales, J. 2009. Review of the *Oryzomys couesi* complex (Rodentia: Cricetidae: Sigmodontinae) in western Mexico (http://digitallibrary.amnh.org/dspace/bitstream/2246/6035/4/331-03-carleton.pdf). *Bulletin of the American Museum of Natural History* 331:94–127.
- Eliot, D.G. 1903. A list of a collection of Mexican mammals with descriptions of some apparently new forms (https://books.google.com/books?id=m3YrAAAAYAAJ). Field Columbian Museum, Zoological Series 3(8):141–149.

- Eliot, D.G. 1904. The land and sea mammals of Middle America and the West Indies (https://books.google.com/books?id=ukEZAAAAYAAJ). Field Columbian Museum, Zoölogical Series 4(1):i–xxi, 1–439.
- Goldman, E.A. 1918. The rice rats of North America (https://books.google.com/books?id=HKYrAAAAYAAJ&pg=PA45). *North American Fauna* 43:1–100.
- Hall, E.R. 1960. *Oryzomys couesi* only subspecifically different from the marsh rice rat, *Oryzomys palustris* (https://www.jstor.org/stable/3669517) (subscription required). *The Southwestern Naturalist* 5(3):171–173.
- Merriam, C.H. 1901. Synopsis of the rice rats (genus *Oryzomys*) of the United States and Mexico (https://books.google.com/books?id=AUYZAAAAYAAJ). *Proceedings of the Washington Academy of Sciences* 3:273–295.
- Musser, G.G. and Carleton, M.D. 2005. Superfamily Muroidea. Pp. 894–1531 in Wilson, D.E. and Reeder, D.M. (eds.). *Mammal Species of the World: a taxonomic and geographic reference*. 3rd ed (http://www.bucknell.edu/msw3). Baltimore: The Johns Hopkins University Press, 2 vols., 2142 pp. ISBN 978-0-8018-8221-0
- Weksler, M. 2006. Phylogenetic relationships of oryzomyine rodents (Muroidea: Sigmodontinae): separate and combined analyses of morphological and molecular data (http://hdl.handle.net/2246/5777). *Bulletin of the American Museum of Natural History* 296:1–149.
- Weksler, M., Percequillo, A.R. and Voss, R.S. 2006. Ten new genera of oryzomyine rodents (Cricetidae: Sigmodontinae) (http://hdl.handle.net/2246/5815). *American Museum Novitates* 3537:1–29.

Retrieved from "https://en.wikipedia.org/w/index.php?title=Oryzomys_albiventer&oldid=1239635604"